21 $f(x) = \dfrac{4}{x+3}$

$g(x) = 3x + 1$

(a) Find $f^{-1}(x)$

$y(x + 3) = 4 \rightarrow x = 4 \div y - 3$

1 mark
Rearrange y = f(x).

$f^{-1}(x) = \dfrac{4}{x} - 3$

1 mark
Interchange x and y.

(2)

Given that $a > 0$

(b) find the set of values of a for which $gf(2a) < a$

$gf(2a) = \dfrac{12}{2a + 3} + 1 < a$

1 mark
Apply f, then g.

Since $a > 0$, $2a + 3 > 0$. Multiply through:

$12 + 2a + 3 < a(2a + 3)$

$2a^2 + a - 15 > 0$

1 mark
Form the quadratic.

$(2a - 5)(a + 3) > 0$

1 mark
Factorise.

Critical values: $a = -3$ and $a = 2.5$.

1 mark
Both critical values.

For $a > 0$, $a + 3 > 0$; hence $2a - 5 > 0$.

Therefore $a > 2.5$.

1 mark
Use the given a > 0.

(5)

(Total for Question 21 is 7 marks)

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